

Numerical Linear Algebra First-Year Exam, August 2024

Do 4 (four) problems.

1. Let A be an $m \times n$ matrix ($m \geq n$) and let $A = \widehat{Q}\widehat{R}$ be a reduced QR factorization of A . Prove that A has full rank if and only if the diagonal entries of $\widehat{R}$ are nonzero.
2. Assume that $A \in \mathbb{C}^{n \times n}$ and there exists $p \geq 1$ such that $\|A\|_p < 1$, where $\|\cdot\|_p$ is a vector-induced matrix norm.

(a) Prove that $I - A$ is invertible.

(b) Prove that

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k.$$

(c) Prove that $\|A\|_q \|A^{-1}\|_q \geq 1$, $\forall 1 \leq q \leq \infty$.

(d) Prove that

$$\frac{1}{1 + \|A\|_p} \leq \|(I - A)^{-1}\|_p \leq \frac{1}{1 - \|A\|_p}.$$

3. Suppose A is Hermitian positive definite.

(a) Prove that each principal submatrix of A is Hermitian positive definite.

(b) Prove that an element of A with largest magnitude lies on the diagonal.

(c) Prove that A has a Cholesky decomposition.

4. Let $\mathbf{v} = [2, -1, 1]^T$ and $H = (\text{span}\{\mathbf{v}\})^\perp$.

(a) Find the matrix $P_{\mathbf{v}}$, the orthogonal projection onto $\text{span}\{\mathbf{v}\}$.

(b) Find the matrix P_H , the orthogonal projection onto H .

(c) Find Q_H , the unitary matrix that reflects across H .

5. Let $A, \delta A \in \mathbb{R}^{n \times n}$ be full rank and b, x and $\delta x \in \mathbb{R}^n$. Prove that if

$$Ax = b, \quad \text{and} \quad (A + \delta A)(x + \delta x) = b,$$

then

$$\frac{\|\delta x\|}{\|x + \delta x\|} \leq \kappa(A) \frac{\|\delta A\|}{\|A\|},$$

where $\kappa(A)$ is the condition number of A .